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A Profile of OxyContin Addiction

Lon R. Hays, MD, MBA

ABSTRACT. OxyContin is a controlled-released form of oxycodone indicated for the management of moderate to severe pain. OxyContin diversion and abuse has become a major problem in certain areas of the United States, particularly rural areas and Appalachia. This study undertakes an 18-month retrospective chart review at a private freestanding psychiatric facility to develop a profile of OxyContin addicts seeking treatment. There were 579 admissions to the Addictive Disease Unit of this facility from October 2000 to March 2002, with 298 of these admissions being for the treatment of opioid abuse or dependence. One hundred and eighty-seven of these individuals were dependent on OxyContin, using an average dose of 184 milligrams of OxyContin per day. The OxyContin-dependent individuals tended to show a progression from p.o. use to either snorting or i.v. use. The author concludes that a socio-cultural understanding is needed to better treat these addicts as is improved communication between the pain treatment community and the addiction treatment community. *[Article copies available for a fee from The Haworth Document Delivery Service: 1-800-HAWORTH. E-mail address: <docdelivery@haworthpress.com> Website: <<http://www.HaworthPress.com>> © 2004 by The Haworth Press, Inc. All rights reserved.]*

KEYWORDS. OxyContin, addiction, oxycodone, pain, opioid

INTRODUCTION

OxyContin is a controlled-release form of oxycodone hydrochloride produced by Purdue Pharma since 1995. This medication is designed to

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provide the delivery of oxycodone over a 12-hour period and is indicated for the management of moderate to severe pain.¹ OxyContin is supplied in 10 mg, 20 mg, 40 mg, and 80 mg tablets. Although all prescription painkillers have the potential for abuse, the recent surge in OxyContin abuse has become epidemic in particular areas of the country. Although it is designed to be swallowed whole, abusers ingest the drug in a variety of ways. OxyContin abusers often chew the tablets or crush the tablets and snort the powder. Because oxycodone is water soluble, the crushed tablets can be dissolved in water and the solution can then be injected. These latter two methods lead to the rapid release and absorption of oxycodone.²

While OxyContin diversion and abuse appears to have begun in the more rural areas of the United States, particularly Appalachia, it has now spread into urban areas. Indeed, its use in rural areas of Appalachia led to the term "Hillbilly Heroin." Other street names are "Oxy," "OC," and "killer." What sets OxyContin abuse apart is the potency of the drug; it contains a much larger amount of the active ingredient, oxycodone, than other prescription opioid pain relievers.³

While concern about prescription drug abuse in our country has risen, there has been a particular concern about the increase in the abuse of prescription painkillers. According to the Drug Abuse Warning Network (DAWN), the incidence of emergency room visits related to narcotic analgesic abuse has been increasing since the mid-1990s and has more than doubled between 1994 and 2001.⁴ Narcotic analgesics were involved in 14% of all drug abuse related emergency room visits in 2001, representing an estimated 90,232 emergency visits; this represents a 117% increase since 1994. The trend for individual narcotic analgesics vary. Those showing the largest increases from 1994 to 2001 included oxycodone (352%), methadone (230%), morphine (210%), and hydrocodone (131%). Although hydrocodone was named in 21,567 visits, oxycodone was second with 18,409 visits. While there was a 41.4% increase from 1999 to 2001 in hydrocodone mentions, there was 186.3% increase in oxycodone mentions.⁴

Many reports of OxyContin abuse and dependence have occurred in rural areas that have housed labor-intensive industries such as logging and coal mining. These industries are often located in economically depressed areas; people for whom the drug may have been legitimately prescribed may be tempted to sell their prescriptions for profit. (A 40 mg pill typically sells for \$40 on the street.) This has led to crime involving robbing pharmacies and writing false prescriptions.³ Pharmacies from Massachusetts to Alaska have been targeted for robberies by

individuals bypassing cash and other narcotics and demanding OxyContin by name.⁵ Some pharmacies in Massachusetts have stopped carrying OxyContin due to the fear of robberies.⁶ In October 2000, 90% of all robberies in Pulaski, Virginia were attributed to OxyContin; and in Hazard, Kentucky over half of the inmates are in jail for OxyContin-related crimes. The state of Maine's Attorney General has called OxyContin abuse, "an addiction epidemic in the state of Maine" where emergency room admissions for OxyContin abuse grew substantially from 1995 to 2001.⁷

In an attempt to decrease abuse by Medicaid patients, Florida, Maine, Ohio, South Carolina, and West Virginia now require prior authorization for OxyContin. Maine requires prior authorization for all patients who do not have cancer. Vermont became the first state to no longer pay for OxyContin for certain recipients of general assistance through the state's social welfare department. Several states have adopted resolutions to study the use and abuse of OxyContin.⁸

The United States DEA has linked OxyContin to 464 overdose deaths in the years 2000 and 2001.⁹ A study by the Florida Department of Law Enforcement showed 218 people died from OxyContin overdoses in the first six-months of 2001.¹⁰ The following states had the most prescriptions of OxyContin per capita in the year 2000: West Virginia, Alaska, Delaware, New Hampshire, Florida, Kentucky, Pennsylvania, Maine, Rhode Island, and Connecticut.¹¹

While there has been a growing concern regarding the abuse of OxyContin, very little information has been published pertaining to the adolescent population. In a Virginia survey of high school students, the majority of students surveyed had heard of OxyContin and felt it was not difficult to obtain. In that particular sample, 9.5% of students claimed to have used OxyContin at least once in their lifetime.¹²

METHODS

The author performed a retrospective chart review from a private freestanding psychiatric facility in Lexington, Kentucky in order to develop a profile of OxyContin addicts seeking treatment. This particular facility is strongly affiliated with the University of Kentucky Medical Center, Department of Psychiatry. The study consists of an 18-month retrospective chart review of adult inpatient cases of OxyContin dependence, as well as other cases of opioid dependence from October 2000 to March 2002. This facility serves patients from the urban Lexington

area as well as from rural areas of Central and Eastern Kentucky. Information for the chart review was obtained from the admission assessment as well as the discharge summary. In addition to demographic data, information was obtained regarding drug of choice, route of administration, duration of use, other drugs of abuse, history of previous treatment, and other Axis I diagnoses.

RESULTS

Over the eighteen-month period, there were a total of 579 admissions to the Adult Addictive Disease Unit. Two hundred and ninety-eight of these admissions (51.3%) were for the treatment of opioid abuse or dependence. One hundred and eighty-seven (62.8%) of these individuals were dependent on OxyContin; 111 (37.2%) were dependent on other opioids. A structured review was performed on the charts representing all individuals who were opioid dependent.

The group of OxyContin-addicted individuals was composed of 135 males (72.2%) and 52 females (27.8%). The mean age of this group was 31.0 years and 91% of these individuals were from rural areas. In terms of other demographics, the mean educational level of the OxyContin addicts was 12.0 years and 51% had children. One hundred and eighty-four of the 187 OxyContin addicted patients admitted to buying their drug off the street, while 50 of the 187 (26.7%) maintained they were also getting prescriptions from physicians.

Ninety (48.2%) of the OxyContin users admitted to abusing other opioids in addition to OxyContin. These included: Lortab (22%); Percocet (13%); Lorcet Plus (12%); Methadone (9%); and Morphine (6%). The initial route of administration of OxyContin was known in 112 of the 187 individuals. Eighty-three percent begin their use by p.o. administration, 16% begin their use by snorting, and only 1% by i.v. use. At the time of admission for treatment, the route of administration was known in 133 individuals. At that time 62.4% were snorting, 25.6% were using i.v., and 14.3% were using p.o. The mean dose of OxyContin used on a daily basis was 184 milligrams. The mean duration of use of OxyContin from the date of first use until treatment was 19.2 months.

Eighty-six of the 187 patients (45.9%) admitted having a pain problem at some time during their OxyContin use, although 20 of these 86 (23.3%) stated they no longer had any type of pain. A review of charts showed that the subjective effect of the drug was known in 49 of the cases. Forty-seven of these 49 individuals described a sense of in-

creased energy, euphoria, invincibility, or “ability to do anything they needed to do.”

Forty-nine percent of the OxyContin addicts were known to have had some type of previous treatment, either inpatient or outpatient. One hundred and twenty-two of the 187 (65.2%) had an Axis I diagnosis in addition to opioid dependence. One hundred and one of these (54% of the total sample) had an additional chemical dependency diagnosis. Forty-three of these diagnoses were sedative hypnotic dependence; 19 were alcohol dependence; 14 were cocaine dependence; 13 were polysubstance dependence; and 11 were cannabis dependence. Fifty-seven of the 122 (30.5% of the total sample) with an additional Axis I diagnosis had a psychiatric diagnosis besides chemical dependency. Twenty-nine had a diagnosis of Major Depression or Depression, NOS; 6 had a diagnosis of Substance-Induced Mood Disorder; 5 had a diagnosis of Mood Disorder, NOS; 5 had a diagnosis of Bipolar Affective Disorder; 4 had Schizoaffective Disorder; and 2 had Panic Disorder.

In addition to the OxyContin addicts, there were 111 individuals with the diagnosis of opioid dependence (besides OxyContin). Fifty-one percent were male and 77% were from rural areas. The mean age of this group was 37.7. The other opioids used by this population were Lortabs (30%), Lorcet or Lorcet Plus (19%), Methadone (11%), Percocet (7%), and Morphine (6%).

During the time period under review, the overall admitted population of the Addictive Disease Unit was 65% rural and 35% urban. One hundred eighty-five individuals were admitted over this period with a primary diagnosis of alcohol dependence; there were 152 males (82.2%) and 33 females (17.8%). The average age of the alcohol-dependent individuals was 43 years. Only 12 individuals were admitted with a primary diagnosis of cocaine dependence. The remainder of admissions were primarily for polysubstance dependence.

DISCUSSION

The number of opioid-addicted individuals at this freestanding psychiatric facility in Lexington, Kentucky increased significantly over the past three years. This 18-month retrospective chart review showed that OxyContin individuals outnumbered all other opioid dependent individuals by a ratio of 5 to 3. The OxyContin addicted individuals were significantly more likely to be male and more likely to be from a rural area than were those dependent on other opioids. The average age of

OxyContin addicts was younger than those addicted to other opioids. They also were more likely to be from a rural area than the other Addictive Disease Unit admissions (91% vs. 65%). Compared to the alcohol-dependent patients, OxyContin addicted patients were younger on average (31 vs. 43). The use of OxyContin in this population demonstrated a progression from p.o. use to either snorting or i.v. use. The fact that 25% of individuals admitted for OxyContin dependence were injecting their drug was quite interesting considering that the area represented is a very "low injecting" area. This has obvious implications for the need to investigate possible medical comorbidities in this population such as HIV, hepatitis, and endocarditis.

The average daily dose of 184 mg certainly represents a "supra-therapeutic" dose of OxyContin. The mean duration of use of OxyContin from date of first use until the time of treatment was 19.2 months. Addiction professionals who have been in the field certainly recognize that this is a very brief time period of drug use necessitating treatment. Part of this relates to the fact that OxyContin is so expensive on the street (\$1.00 per milligram), meaning that individuals had often exhausted all of their financial resources requiring them to come into treatment, or face marked opioid withdrawal on their own.

In addition to relieving pain, opioid drugs affect particular regions of the brain that mediate pleasure, possibly resulting in initial euphoria. One of the most interesting parts of this chart review had to do with the sense of increased energy and euphoria as well as invincibility described by users. (This sounds very much like the subjective effect described by cocaine addicts.) The two brain regions often identified as mediating reinforcement properties of drugs are the ventral tegmentum and the nucleus accumbens. In animal models, direct stimulation of the nucleus accumbens and the injection of dopamine agonists into this region are reinforcing. Mu opioid agonists increase activity of dopaminergic neurons originating in the ventral tegmentum by inhibiting the GABA interneurons.¹³ It could be this dopaminergic stimulation that is causing the euphoric response from such a significant dose of opioid agonist.

Treatment professionals working in Appalachian areas need to be aware that they are facing conditions stemming from the region's culture and history. The Lexington Herald-Leader reported that in parts of six counties in Eastern Kentucky, drugstores, hospitals and other dispensaries received more narcotics per person than anywhere else in the nation over the past four years. A history of prescription drug use, tied in part to physically demanding industries such as logging and mining, created an acceptance of pills that affects how some people regard abuse

of medications. The historic isolation of the region led to a strong reliance on self and family, which often means that people wait longer to seek help and exhaust family resources, resulting in them being sicker when they finally do seek professional help.¹⁴

Some of the strongest arguments against chronic opioid therapy concern psychiatric diagnoses that are important factors to consider. While it is obviously inappropriate to use opioids to treat depression and anxiety, it is also clear that opioids can worsen depression and that opioid micro-withdrawal often produces anxiety and irritability. There are also studies suggesting that psychological function worsens with chronic opioid therapy.¹⁵ In this particular retrospective chart review there were a large number of OxyContin dependent patients who had diagnoses of depression. It was felt by the author that a number of these diagnoses could have been substance-induced depression had the length of stay been longer than an average of 5-7 days.

The increase in abuse of OxyContin has resulted in a significant increase of patients admitted to methadone treatment programs. Programs in West Virginia, Pennsylvania, Kentucky, and Virginia report that 50-90% of newly admitted patients identified OxyContin as their primary drug of abuse. One treatment facility in Virginia reported that of 291 clients currently enrolled in their program, 240 are being treated for OxyContin abuse.⁵

CONCLUSION

When attempting to understand the OxyContin phenomena in Eastern Kentucky and other rural and Appalachian areas, it is necessary to consider a thoughtful sociological analysis. It is necessary to understand communities marked by poverty and significant unemployment among laborers with significant amounts of chronic pain. There needs to be good long-term research on outcomes and investigators committed to carrying it out.¹⁶

Kentucky is 1 of 18 states with an automated database that pharmacists, physicians and law enforcement officials can use to track whether a person is filling multiple prescriptions for the same drug at various locations. This system helps to identify people who are obtaining high volumes of a particular drug. All pharmacies including mail order pharmacy programs that ship drugs into Kentucky must report information to the state.¹⁷ Legal authorities feel this program is helping and the state has increased its funding for the system.

Healthcare providers must acknowledge that the properties of this drug that make it potentially addictive and dangerous for recreational users also make it effective for patients with cancer who experience chronic pain. Therefore, healthcare providers need to work to protect its use while preventing its abuse.¹⁸ Although it is often believed that longer acting drugs have a lesser risk of abuse, this concept has sometimes been overstated. It is helpful to look at the treatment of pain on a continuum. On one end of that continuum lie individuals being treated for acute pain. On the other end of that continuum would be long-term opioid treatment for pain in an addict population. The risk of aberrant drug related behavior at this end of the spectrum can be quite high. Physicians must be taught to assess where a patient falls along this continuum of addiction risk. Furthermore, they must also be able to assess an individual's psychosocial condition and subculture well enough to judge the addiction risk for the particular individual.¹⁹

Healthcare providers must not ignore the risk of addiction and must assess and reassess patients carefully looking not only at a history of substance abuse, but also at psychological and social conditions that can contribute to risk. Patients with a history of chemical dependency should be treated with structures that encourage close adherence to a treatment plan and impede abuse and diversion. There certainly is a shortage of settings that can treat pain and bring the needed amount of structure when needed. Much more cooperation is needed between the pain management and chemical dependency communities to develop and support these settings.¹⁹

Healthcare providers must assume responsibility for education regarding the abuse and diversion of prescription drugs. Healthcare providers, pharmacies, and manufacturers must continue to improve efforts in pain management, but at the same time use extreme care to reduce the diversion of opioid medications for non-medical use.

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